# Balla Pavan Kumar

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## Education

**National Institute of Technology, Andhra Pradesh** **2022 – 2026**
*Bachelor of Technology in Electrical and Electronics Engineering - CGPA: 8.28/10.00* *Tadepalligudem, Andhra Pradesh*

**Narayana Junior College** **2019 – 2021**
*Aggregate: 96.6%* *Vijayawada, Andhra Pradesh*

**Narayana E-techno School** **2018 – 2019**
*Aggregate: 10 CGPA* *Kakinada, Andhra Pradesh*

## Relevant Coursework

- Video editing
- Content writing
- Python
- Machine Learning
- Deep Learning

## Technical Skills

**Languages**: C++, HTML, CSS, Python, SQL
**Developer Tools**: GitHub, VS Code, Microsoft Clipchamp, Adobe Premiere Pro
**Core Skills**: Machine Learning, Deep Learning, NLP, video Editing, Content writing
**Libraries/Frameworks**: Scikit-learn, Pandas, NumPy, Flask

## Projects

**Mini RAG Chatbot** | *Python, FAISS, Sentence Transformers, OpenAI GPT-4o-mini, NLP, Vector Embeddings, Flask* |

- Developed a Retrieval-Augmented Generation (RAG) chatbot using Python, FAISS, Sentence Transformers, and OpenAI GPT-4o-mini to retrieve semantically relevant document chunks and generate accurate, grounded responses from internal documents.
- Implemented document chunking, vector embeddings with all-MiniLM-L6-v2, and FAISS-based similarity search to improve retrieval accuracy, reduce hallucinations, and enhance transparency by displaying retrieved context alongside generated answers.

**House Price Prediction using Machine Learning** | *Python, Flask, Scikit-learn, Pandas, NumPy, HTML, Joblib* |

- Developed a Machine Learning-based house price prediction web application using Python and Flask to estimate property prices based on features such as area, bedrooms, bathrooms, parking, and furnishing status.
- Built and trained a Random Forest Regression model using Scikit-learn with data preprocessing, feature encoding, and model integration through Joblib to enable accurate real-time predictions via an interactive web interface.

**Muti-Mode Autonomous Robot** | *Arduino UNO, IR Sensors, Ultrasonic Sensors, HC-05 Bluetooth Module, App Inventor*

- Designed and developed a multi-mode autonomous robot capable of line following, obstacle avoidance, Bluetooth-based manual control, and voice command operation using Arduino UNO and sensor integration.
- Implemented autonomous navigation and real-time obstacle detection using IR and ultrasonic sensors, while enabling remote control through an Android application using the HC-05 Bluetooth module.

## Achievements / Extracurricular

- Received a "Certificate of Appreciation" for presenting the MultiMode Autonomous Robot at GENESIS-2025 at NIT Andhra Pradesh .
- Earned NPTEL Elite + Top 1% certification with a score of 96% in "Introduction to Internet of Things" by IIT Kharagpur .
- Certificate for "Fundamentals of Deep Learning"– NVIDIA .
- Executive Member, Resource Hub — Assisted in sharing academic resources and guidance for juniors.
- Executive Member, Hospitality Club — Managed 300+ guests during TECHKRIYA'25, and VULCANZY'25..
- Executive Member, E-Yantra Club — Participated in robotics projects.